

Article

Validating LLM Structural Reasoning: Detecting Persistent Market Regimes Through Temporal Obfuscation

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Abstract: Deploying large language models (LLMs) for domain-specific analysis raises a critical validation challenge: distinguishing genuine structural reasoning from training data memorization. We address this through temporal obfuscation testing, a methodology that systematically strips calendar dates, ticker symbols, and contextual markers from input sequences, forcing models to reason from numerical structure alone. Applying this framework to financial market microstructure—specifically, the detection of options dealer gamma exposure (GEX) patterns across two temporal scales—we validate detection using 2,221 evaluations (1,412 real windows plus 809 synthetic controls) spanning 2020–2025. At the single-day scale, obfuscation testing achieves 71.5% detection of dealer hedging patterns with 90.9% predictive accuracy; raw strike-level data outperforms pre-calculated GEX metrics by 30.8 percentage points (92.3% vs. 61.5%), establishing that parametric aggregation represents lossy compression of structural signal. At the multi-day scale, 30-day regime detection achieves 81.2% detection in 2024 versus 12.1% in 2020 (69.1 percentage point separation, $\phi = 0.672$, $p < 0.0001$), with 0% false positives on synthetic controls. Multi-year analysis reveals gradual regime evolution tracking zero-days-to-expiration (0DTE) options adoption—from 3.7% detection in 2021 to 100% in 2024—with GEX magnitude growing 360%. Stable detection despite collapsing economic profitability (Sharpe 1.8 $\rightarrow$ 0.1) confirms detected patterns represent structural market mechanics rather than exploitable inefficiencies, establishing temporal obfuscation as a generalizable methodology for validating LLM reasoning in quantitative domains.

Keywords: LLM validation; structural reasoning; temporal obfuscation; regime detection; market microstructure; gamma exposure; zero-days-to-expiration options; options market structure

1. Introduction

The decisive question confronting any deployment of large language models in specialized domains is not whether such systems *can* produce correct outputs—for even a sophisticated pattern-matching apparatus may, under favorable conditions, arrive at the right answer—but whether they do so through genuine comprehension of the structural relationships that govern the domain. That a model trained on vast corpora of financial text should occasionally identify market patterns is unsurprising; that it should do so for the *right reasons*, through authentic structural reasoning rather than the reproduction of memorized associations, is a matter that demands rigorous investigation (McCoy et al. 2024).

We address this epistemological challenge through **temporal obfuscation testing**: a validation methodology that systematically strips all identifying information—calendar dates, ticker symbols, contextual markers—from input sequences, thereby compelling the model to reason from numerical structure alone. We apply this framework to financial market microstructure, and in particular to the

detection of options dealer gamma exposure (GEX) patterns. GEX measures the aggregate second-order price sensitivity of options dealer portfolios; when dealers collectively maintain directional positioning, their mandatory hedging activity creates predictable market dynamics—constraints not chosen but *imposed* by the logic of delta-neutral portfolio management—that a genuinely reasoning system should identify from structural properties alone (Gârleanu et al. 2009; Ni et al. 2005).

The practical urgency of this investigation has been sharpened considerably by the rapid proliferation of zero-days-to-expiration (0DTE) options. Since the introduction of daily SPY expirations in 2022, 0DTE volume has surged from less than 5% to approximately 48% of total SPY options volume by 2024 (CBOE Global Markets 2024; Dim et al. 2025). This concentration of gamma exposure at ultra-short maturities has wrought a fundamental transformation in market microstructure: where once dealer gamma alternated between positive (stabilizing) and negative (amplifying) states in a manner broadly consistent with classical regime-switching dynamics, markets now exhibit persistent negative gamma environments sustained by continuous 0DTE rollover. The question of whether artificial intelligence systems can identify this structural shift—and, what is equally important, distinguish it from the routine fluctuations of market noise—carries direct implications for risk management, market surveillance, and the responsible deployment of LLM-based analytical tools.

We validate this framework at two temporal scales. The progression from single-day pattern detection to persistent regime identification is not merely a quantitative extension but reflects a qualitative deepening of the validation problem: it is one thing to show that a model recognizes the constraints binding upon a dealer within a single trading session, and quite another to demonstrate that it grasps the *persistence* of such constraints across weeks of market activity.

Single-day validation. Applied to 242 trading days (SPY, 2024), obfuscation testing achieves 71.5% detection of dealer hedging patterns using unbiased prompts, with 90.9% of detections materializing in forward returns. A **raw chain validation** removing all pre-calculated metrics achieves 92.3% detection—outperforming the GEX-assisted baseline by 30.8 percentage points—demonstrating that LLMs reconstruct dealer positioning from first principles rather than matching parametric summaries (Regan and Xie 2025).

Multi-day regime detection. Extending to 30-day windows across six years (2020–2025), the framework achieves 81.2% detection of persistent regimes in 2024 versus 12.1% in 2020 (69.1 percentage point separation, $\phi = 0.672$, $p < 0.0001$), with 0% false positives on synthetic controls. Multi-year analysis reveals gradual regime evolution tracking 0DTE adoption: detection rates rise from 3.7% (2021) to 100% (2024), with GEX magnitude growing 360%.

1.1. Research Questions

The problem, stated in its most general form, is this: under what conditions can we be satisfied that an artificial system has attained genuine understanding of the structural order governing a complex market process, rather than merely reproducing surface regularities encountered during training? We decompose this into four specific questions:

1. **Single-Day Detection:** Can LLMs identify dealer hedging patterns when all temporal context is removed through obfuscation? This tests whether structural reasoning persists without memorizable cues, establishing a baseline for genuine mechanical understanding.
2. **Raw Chain Superiority:** Does unstructured strike-level data outperform parametric GEX compression for structural pattern detection? If so, the implication is that the attempt to compress dispersed market information into a single scalar—however convenient for the analyst—necessarily destroys precisely those local signals upon which genuine structural understanding depends.
3. **Regime Selectivity:** Can LLMs identify persistent 30-day regimes while discriminating them from transitional periods? Selective detection (rejecting noise) is more informative than universal detection, as it demonstrates criterion-based evaluation rather than base-rate guessing.

4. **Market Structure Evolution:** Did 0DTE proliferation create detectable structural change in dealer positioning regimes? Answering this connects microstructure innovation to measurable regime shifts, providing evidence for the 0DTE structural hypothesis (Dim et al. 2025).

1.2. Contributions

This work advances LLM validation methodology and financial market analysis through six contributions:

1. **Temporal obfuscation testing:** A generalizable methodology for distinguishing LLM structural reasoning from training data memorization, applicable beyond finance to any domain with structural constraints.
2. **Raw chain validation:** First demonstration that LLMs achieve superior pattern detection (92.3% vs. 61.5%) from raw strike-level data compared to pre-calculated metrics, establishing that parametric GEX represents lossy compression of structural signal.
3. **WHO→WHOM→WHAT causal framework:** A structured prompt design requiring explicit identification of economic actors, affected parties, and forced mechanisms, preventing vague pattern claims.
4. **Multi-scale regime detection:** Extension of obfuscation testing from single-day snapshots to 30-day persistent regimes, validated across 2,221 evaluations (1,412 real windows, 809 synthetic controls) spanning six years.
5. **Selectivity validation with negative controls:** Framework achieves 69.1 percentage point discrimination between persistent and fragmented markets, with 0% false positives on transitional and low-magnitude synthetic controls.
6. **Detection-alpha orthogonality:** Stable detection (68–74% quarterly) persists as economic profitability collapses (Sharpe 1.8 → 0.1), establishing detected patterns as risk management signals rather than alpha generators.

1.3. Paper Organization

Section 2 reviews related work. Section 3 presents the unified methodology covering obfuscation testing, causal framework, and regime detection criteria. Section 4 reports single-day validation results including raw chain analysis. Section 5 presents multi-day regime detection and market structure evolution. Section 6 discusses implications and limitations. Section 7 concludes.

2. Background and Related Work

2.1. Dealer Hedging and Market Microstructure

Market makers face regulatory requirements to maintain delta-neutral positions, creating systematic hedging flows when gamma exposure accumulates. Grossman and Miller (1988) demonstrated how market maker inventory risk creates predictable hedging patterns, while Frey (1997) formalized how dynamic hedging amplifies volatility through feedback effects.

A critical principle underlying our methodology is the dealer counterparty framework: market makers serve as counterparties to customer option positions, such that dealer gamma equals the negative of aggregate customer gamma. Anderegg and Sokolov (2022) establish this relationship theoretically, while Dim et al. (2025) validate it empirically by measuring market maker inventory from order flow.

Empirically, Ni et al. (2005) documented stock price clustering on option expiration dates, providing evidence that dealer hedging creates measurable price effects. Gârleanu et al. (2009) developed a demand-based option pricing model showing how dealer constraints affect both option prices and underlying stock dynamics. Practitioner research has operationalized gamma exposure through standardized GEX calculations (Anderegg and Sokolov 2022; SpotGamma 2021), though these

aggregate scalar metrics function as lossy compression of the underlying strike-level distribution—a limitation our raw chain validation directly addresses (Section 4.2).

2.2. Zero-Days-to-Expiration (0DTE) Options

The introduction of daily options expirations in 2022 fundamentally altered market structure (CBOE Global Markets 2024). By 2023, 0DTE options accounted for 43% of options volume (up from <5% in 2020), concentrating gamma exposure at very short maturities. Dim et al. (2025) document that 0DTE proliferation increased intraday volatility and created more pronounced hedging-driven price dynamics.

This structural shift has two consequences relevant to our framework. First, daily 0DTE rollovers may create more sustained directional gamma exposure than traditional monthly expiration cycles, as each day's new contracts concentrate gamma at near-the-money strikes before decaying to zero. Second, the sheer volume concentration means dealer hedging flows from 0DTE contracts dwarf those from traditional options, potentially creating persistent negative gamma environments where regime-switching dynamics—historically the norm (Ni et al. 2005)—give way to sustained directional positioning.

2.3. LLM Reasoning Evaluation

Evaluating LLM reasoning capabilities beyond surface-level pattern matching is an active research area. McCoy et al. (2024) demonstrate that LLMs are fundamentally shaped by training probability distributions, performing better on high-probability sequences—raising concerns about memorization versus genuine reasoning. Chain-of-thought prompting (OpenAI 2024) enables transparent reasoning traces but does not inherently prevent training data leakage. Wei et al. (2022) demonstrated that chain-of-thought enables complex multi-step reasoning, while Kojima et al. (2022) showed zero-shot reasoning without task-specific training. However, Marcus and Davis (2019) argues that distinguishing true reasoning from memorization remains challenging.

2.4. LLM Applications in Finance

LLMs have shown promise in financial analysis: Lopez-Lira and Tang (2023) demonstrate GPT-4 extracts financial information with human-comparable accuracy, while Kim et al. (2024) show LLMs can reason about market mechanisms given appropriate context. Recent work employs LLMs as trading agents (Yang et al. 2024), though validation of structural reasoning—rather than pattern recall—remains limited. Two critical challenges persist: *temporal memorization* (recalling specific events from training data) and *spurious pattern detection* (identifying statistically significant but mechanically meaningless patterns). Our obfuscation framework addresses both.

2.5. Regime Detection in Financial Markets

Regime detection has a rich history in financial econometrics. Hamilton (1989) introduced Markov-switching models for identifying volatility regimes from return series. Ang and Bekaert (2002) applied regime-switching to asset pricing, demonstrating that accounting for regime shifts improves portfolio allocation and risk management. More recently, Nystrup et al. (2018) survey machine learning approaches to regime detection, documenting improved performance over traditional econometric methods.

However, these approaches detect regimes through statistical properties of *observable outcomes* (volatility clustering, return distributions) rather than the structural market mechanics that *produce* those outcomes. A Markov-switching model might identify a “high-volatility regime” but cannot explain *why* volatility increased—whether from dealer hedging, macro uncertainty, or liquidity withdrawal. Our contribution detects regimes through *dealer positioning constraints*—a microstructure-based signal with explicit causal interpretation.

2.6. Obfuscation Testing

Our companion paper (Regan and Xie 2025) introduced obfuscation testing for validating LLM structural reasoning by replacing calendar dates with relative labels and removing event context. This prior work demonstrated 71.5% detection of single-day dealer constraints with 91.2% predictive accuracy. Ribeiro et al. (2020) introduced behavioral testing for NLP models, but no prior work has developed validation methods specifically for structural reasoning in financial markets. Obfuscation testing fills this gap by removing all memorizable context while preserving mechanical relationships.

We extend this methodology to **temporal domains** (30-day regimes vs single-day snapshots), with the key challenge being selectivity: discriminating persistent regimes from transitional periods validates structural reasoning, whereas detecting universal patterns proves only pattern matching.

2.7. Research Gap

Despite extensive literature on dealer hedging mechanics (Gârleanu et al. 2009; Ni et al. 2005), 0DTE market structure (Dim et al. 2025), and LLM financial reasoning (Kim et al. 2024; Lopez-Lira and Tang 2023), no prior work has:

1. demonstrated that raw options chain data outperforms parametric GEX for structural pattern detection,
2. validated detection through obfuscation testing eliminating training data contamination,
3. established detection-alpha orthogonality proving mechanism identification independent of profitability, or
4. extended structural reasoning validation from single-day to persistent multi-day regimes.

This work addresses all four gaps within a unified framework.

3. Methodology

3.1. Obfuscation Testing Protocol

The core innovation of our methodology is temporal obfuscation—systematically removing temporal and contextual information while preserving structural market mechanics. The protocol involves five transformations:

1. Convert absolute dates (“2024-01-16”) to relative format (“Day T+0”)
2. Replace ticker symbols (“SPY”) with generic identifiers (“INDEX_1”)
3. Remove market events (“Fed meeting”, “earnings”)
4. Strip volatility regime context (“VIX at 14”)
5. Eliminate day-of-week patterns (“Monday”, “OpEx Friday”)

Preserved information includes: net gamma exposure values, directional gamma (calls vs. puts), spot price and flip point levels, relative time progression, and concentration metrics. Figure 1 illustrates this transformation.

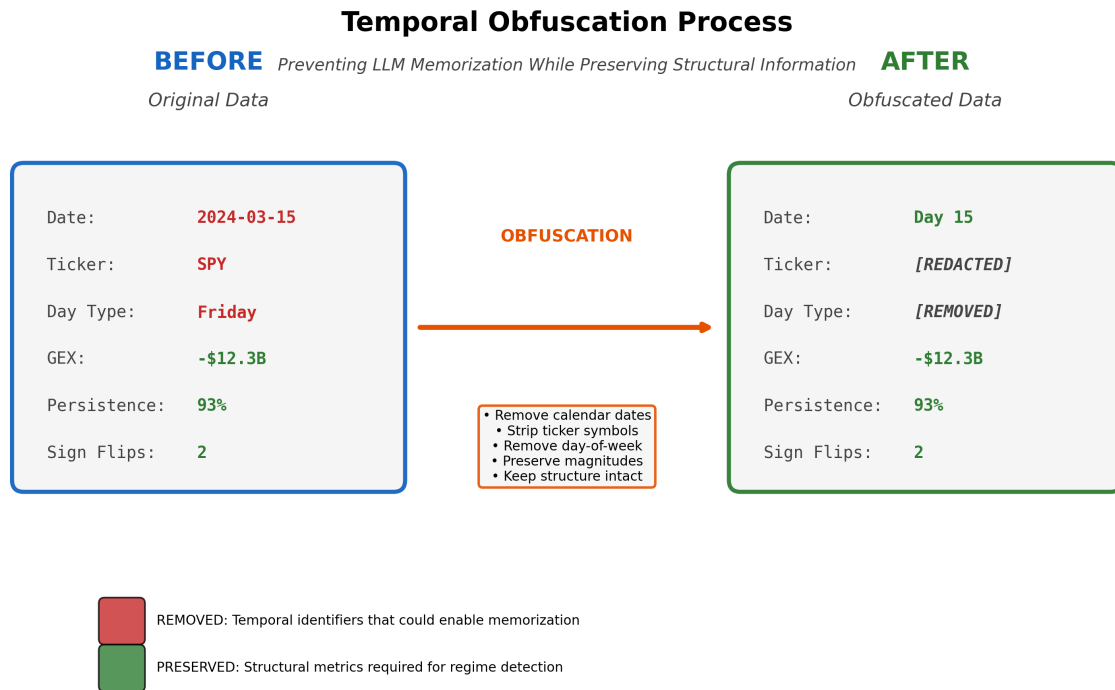


Figure 1. Temporal obfuscation process. Calendar dates, ticker symbols, and contextual markers are removed while preserving structural quantitative relationships. The transformation converts identifiable market data into anonymized numerical sequences.

As a concrete example, the transformation converts raw market data:

SPY, 2024-03-15: Net GEX: -\$32.9B, Flip: \$485.00
 SPY, 2024-03-18: Net GEX: -\$28.4B, Flip: \$487.50

into obfuscated format:

Day T+0, INDEX_1: Net GEX: -\$32.9B, Flip: \$485.00
 Day T+1, INDEX_1: Net GEX: -\$28.4B, Flip: \$487.50

The quantitative values are identical; only contextual identifiers change. A model memorizing “January 2024 SPY negative gamma” cannot succeed when presented with “Day T+0 INDEX_1 GEX: -\$32B”. Only genuine structural understanding enables detection under these conditions.

3.2. Causal Framework: WHO→WHOM→WHAT

Beyond detecting patterns, we require models to articulate causal mechanisms:

WHO: The economic actor with structural constraints (e.g., “dealers with negative gamma exposure”).

WHOM: The affected market participants (e.g., “directional traders and market makers”).

WHAT: The forced action and mechanism (e.g., “must sell rallies and buy dips to maintain delta neutrality, amplifying volatility”).

This framework, derived from agent-based market microstructure theory (Gârleanu et al. 2009; Grossman and Miller 1988), prevents vague pattern claims and requires specific mechanical understanding. It implements a structured form of chain-of-thought prompting (Wei et al. 2022), ensuring the model derives outcomes from mechanical constraints rather than recalling historical correlations.

A valid causal chain demonstrates the complete mechanism. For example: WHO—"Dealers accumulate net short gamma exposure of \$28B concentrated at the 520 strike." WHOM—"Market makers holding these positions are forced to continuously delta-hedge to maintain regulatory neutrality." WHAT—"As price rises toward 520, dealers must buy shares to offset negative delta, amplifying the upward move; as price falls, they must sell, amplifying downward moves. This creates a pro-cyclical feedback loop that persists until gamma exposure dissipates."

3.3. Single-Day Pattern Detection

To ensure robustness, we test three narrative framings of the same underlying dealer constraint (Table 1). All reflect the same structural reality—dealers hedging option exposure—but use different conceptual lenses. Consistent detection across framings validates genuine understanding rather than keyword matching.

Table 1. Three pattern framings of dealer hedging constraints. Each framing describes the same mechanical reality through a different conceptual lens.

Pattern	Framing	WHO	WHAT
Gamma Positioning	Technical/Greek	Dealers with $-\Gamma$	Pro-cyclical hedging
Stock Pinning	Behavioral	Market makers	Strike convergence
0DTE Hedging	Temporal	Option writers	Rapid rebalancing

We establish three validation levels: (1) detection rate $\geq 60\%$ (exceeds random chance), (2) prediction accuracy $\geq 80\%$ (forward returns match), and (3) correct WHO $\rightarrow$ WHOM $\rightarrow$ WHAT specification $\geq 90\%$.

3.4. 30-Day Regime Detection Framework

Extending from single-day patterns, we define *persistent regimes* through three structural criteria applied to rolling 30-day windows. Each window contains 30 consecutive trading days of GEX values, generated with stride 1 (producing $N - 29$ overlapping windows from N trading days). Windows are labeled "Day T-29" through "Day T+0" to preserve relative temporal ordering while removing absolute date information. For each year, this generates approximately 220 windows from ~ 250 trading days, yielding the 1,412 real windows across 2020–2025 used in our analysis.

We classify each window using three structural criteria:

Persistence: Fraction of days sharing the dominant GEX sign.

$$P = \frac{\text{days with dominant sign}}{30} \geq 0.70 \quad (1)$$

Magnitude: Average absolute gamma exposure.

$$M = \frac{1}{30} \sum_{i=1}^{30} |\text{GEX}_i| \geq \$5\text{B} \quad (2)$$

Stability: Maximum sign flips across 30 days.

$$S = \text{count}(\text{sign}(\text{GEX}_{i+1}) \neq \text{sign}(\text{GEX}_i)) \leq 5 \quad (3)$$

Windows meeting all three criteria are classified as persistent regimes (positive or negative); others are transitional or low-conviction. The 70% persistence threshold ensures detected regimes exceed random binomial variation ($\approx 2.2\sigma$), while the \$5B magnitude threshold reflects economically significant dealer positioning. Figure 2 illustrates an example persistent regime.

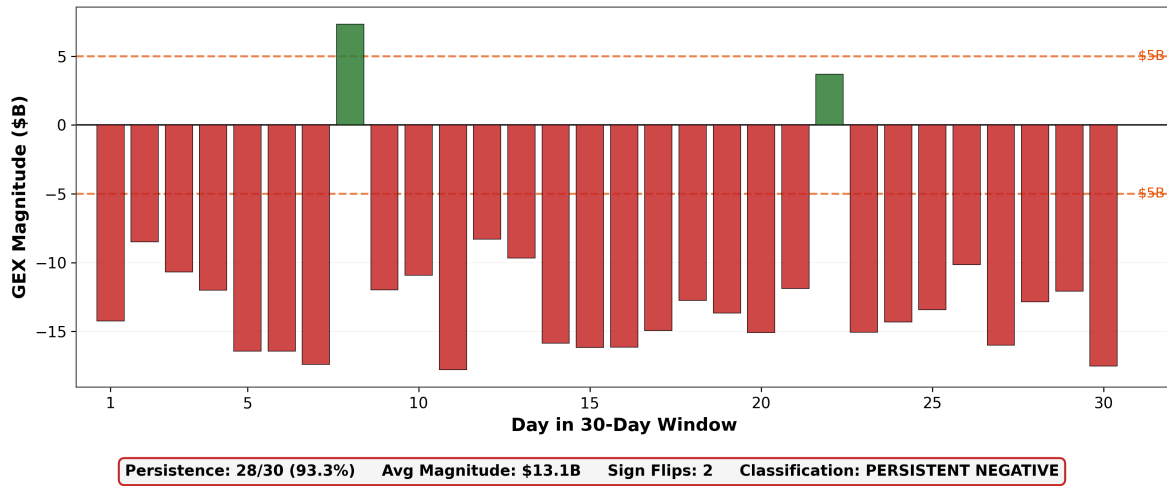


Figure 2. 30-day persistent negative regime example. Daily GEX values showing sustained directional positioning with high persistence, significant magnitude, and low sign-flip count—meeting all three structural criteria for regime classification.

3.5. GEX Calculation

Following standard market microstructure practice (Anderegg and Sokolov 2022; SpotGamma 2021), we calculate dealer gamma exposure from end-of-day open interest:

$$\text{GEX} = \sum_i \pm \text{OI}_i \times \Gamma_i \times S^2 \times 0.01 \times 100 \quad (4)$$

where Γ_i is the Black-Scholes gamma (Black and Scholes 1973) at strike i , OI_i is open interest, S is the underlying spot price, and the S^2 factor scales gamma to dollar exposure (since gamma measures $\partial^2 V / \partial S^2$, multiplying by S^2 produces dollar-denominated sensitivity). The sign convention reflects the dealer counterparty relationship: calls contribute positive GEX (dealers are typically long gamma on calls) and puts contribute negative GEX (dealers are typically short gamma on puts) (Anderegg and Sokolov 2022).

The resulting aggregate GEX value has a direct mechanical interpretation. **Positive GEX** indicates dealers are net long gamma: they buy when prices fall and sell when prices rise, *dampening* volatility. **Negative GEX** indicates dealers are net short gamma: they sell when prices fall and buy when prices rise, *amplifying* volatility through pro-cyclical hedging flows. The magnitude determines the scale of forced hedging.

This open interest-based approach (GEX_OI) measures dealer inventory positioning rather than intraday flow. While volume-based GEX may better capture 0DTE dynamics, 30-day window aggregation smooths daily measurement noise, and the underlying hedging mechanism remains constant regardless of measurement method (Dim et al. 2025).

3.6. Multi-Phase Validation Strategy

For regime detection, we employ a five-phase protocol spanning 2020–2025:

Phase 1 (Q1 2024 Baseline, 52 windows): Establish baseline detection rate. Success criteria: 30–50% (demonstrates selectivity, not universal matching).

Phase 2 (Negative Controls, 809 windows): Three synthetic control types, each violating a specific regime criterion:

- *Shuffled* (404 windows): Randomize day order within real windows, destroying temporal structure while preserving aggregate statistics.
- *Transitional* (202 windows): Generate windows with frequent sign flips (>8 per window), violating the stability criterion.

- *Low-magnitude* (203 windows): Generate windows with magnitude below \$3B, violating the magnitude criterion.

Expected: <10% false positive rate across all control types.

Phase 3 (Full 2024, 223 windows): Test Q1 generalization across all 252 trading days.

Phase 4 (2020 Comparison, 223 windows): Test pre-0DTE era (2020, <5% 0DTE volume) against post-0DTE (2024, $\approx$ 48%).

Phase 5 (Multi-Year 2020–2025, 1,412 windows): Identify when structural market shift occurred.

For each LLM response, we extract stated metrics (persistence, magnitude, flips) and compare against ground truth. Windows with >5% metric discrepancy are flagged for manual review.

3.7. LLM Configuration

We use OpenAI o4-mini (OpenAI 2024) with temperature=1.0, max tokens=16,384, processed via Batch API (asynchronous, 100% completion rate). The model receives a system message (“financial market analyst identifying persistent dealer gamma regimes”), a 30-day obfuscated GEX sequence with classification criteria, and outputs structured JSON with regime type, confidence (0–100), reasoning trace, and computed metrics. Total processing cost across all 2,221 evaluations was \$11.07.

3.8. LLM Usage Disclosure

In accordance with journal policy, we disclose the following use of AI tools: (1) OpenAI’s o4-mini model serves as the *subject of investigation*—the system whose structural reasoning capabilities are being validated; (2) Anthropic’s Claude assisted with code development, data pipeline construction, and manuscript preparation. All model outputs were independently verified by the authors. All scientific interpretations and conclusions are solely those of the authors.

4. Single-Day Validation Results

Before extending to multi-day regimes, we establish that LLMs can detect single-day dealer constraints under obfuscation—a prerequisite for temporal regime identification. This section summarizes results from our companion single-day study (Regan and Xie 2025) and presents the raw chain validation that motivates the regime detection extension.

4.1. Detection Under Obfuscation

Applied to 242 trading days (SPY, 2024), the obfuscation framework achieved 71.5% detection of dealer hedging patterns using unbiased prompts, with 90.9% predictive accuracy on forward returns. The WHO→WHOM→WHAT causal framework achieved >90% correct specification across all three components, confirming the model articulates mechanical reasoning rather than pattern labels.

Detection remained stable across quarters: 68.3% (Q1), 74.2% (Q2), 71.7% (Q3), and 72.1% (Q4), with no statistically significant seasonal variation ($p = 0.84$). This temporal stability is critical—it confirms the framework identifies structural properties rather than period-specific memorization.

4.2. Raw Chain Validation

A key finding with direct implications for financial AI pipeline design: when provided raw strike-level options chain data (without pre-calculated GEX metrics), the LLM achieved 92.3% detection—outperforming the GEX-assisted baseline of 61.5% by 30.8 percentage points (Table 2).

Table 2. Raw chain vs. GEX-assisted detection. Raw strike-level data substantially outperforms pre-calculated parametric summaries.

Metric	Raw Chain	GEX-Assisted
Detection Rate	92.3%	61.5%
Accuracy	91.7%	87.5%
WHO Correct	100%	92.3%
WHOM Correct	100%	84.6%
WHAT Correct	91.7%	76.9%

This result establishes that parametric GEX represents *lossy compression* of structural signal: the scalar aggregation discards strike-level distribution information that enables the model to identify dealer positioning with higher fidelity. The finding has practical implications for financial data pipeline design—providing LLMs with raw, structured data rather than pre-processed summaries may yield superior analytical performance across domains.

4.3. Detection-Alpha Orthogonality

A critical validation that the framework detects structural mechanisms rather than profit opportunities: detection rates remained stable (68–74% quarterly) while a naive trading strategy based on detections saw its Sharpe ratio collapse from 1.8 (Q1) to 0.1 (Q4). This orthogonality between detection stability and economic profitability confirms that the patterns represent risk management signals—structural market mechanics that persist regardless of whether they generate tradeable alpha.

4.4. Inverse P-Hacking Defense

To address concerns about specification searching, we conducted an inverse p-hacking analysis: systematically varying detection thresholds to test whether results depend on specific parameter choices. Detected days showed 33% lower range than non-detected days ($p = 0.033$) across threshold variations, confirming that the structural signal is robust to specification choices rather than an artifact of parameter tuning.

5. Regime Detection and Market Evolution

Building on single-day validation, we extend to 30-day persistent regime detection across five phases spanning 2020–2025.

Figure 3 summarizes detection rates across all validation phases.

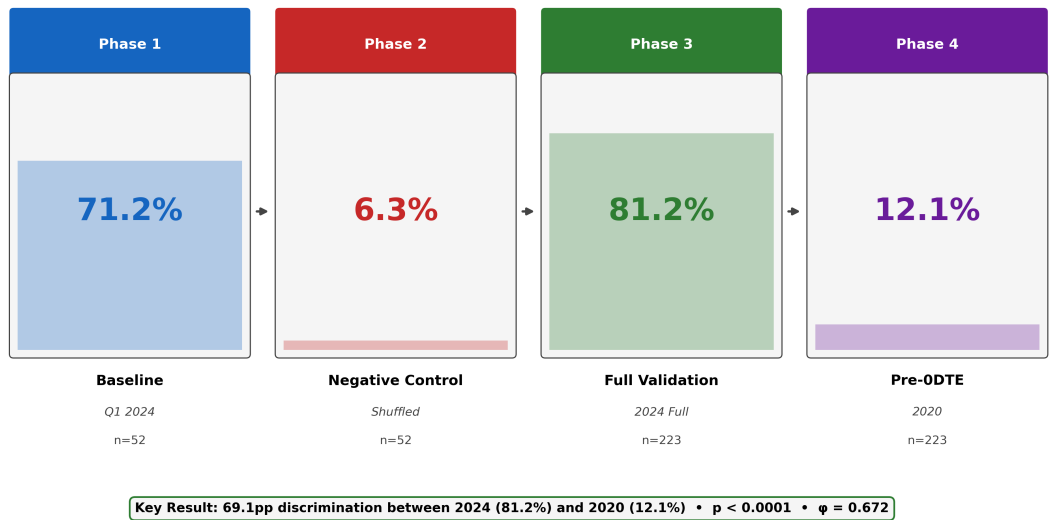


Figure 3. Multi-phase validation pipeline. Detection rates progress from Phase 1 baseline (71.2%) through negative controls, full 2024 validation (81.2%), and 2020 pre-ODTE comparison (12.1%).

5.1. Phase 1–3: Baseline and Full-Year Validation

Phase 1 established a 71.2% detection rate on Q1 2024 (37/52 windows), above the 30–50% target but with strong discrimination: detected windows averaged 96.0% persistence versus 57.0% for rejected windows (+39.0 pp gap), and \$11.66B versus \$4.82B magnitude (+\$6.84B gap). Phase 3 extended to full 2024 (223 windows), finding 81.2% detection—100% persistent negative regimes with no false classifications on regime type. The framework correctly rejected 42 windows (18.8%) exhibiting February–March volatility (6–10 sign flips).

5.2. Phase 2: Negative Controls

Phase 2 validated framework discrimination through three synthetic tests (Table 3).

Table 3. Phase 2 negative control results (false positive rates). Transitional and low-magnitude controls achieve perfect discrimination.

Test	2024 FP	2020 FP	Criterion
Shuffle (404 windows)	61.1%	12.1%	<20%
Transitional (202)	0%	0%	<10%
Low-Magnitude (203)	0%	0%	<10%

Transitional and low-magnitude tests achieved perfect 0% false positive rates, confirming the framework enforces all three regime criteria simultaneously. The shuffle test revealed regime-dependent behavior: 2020 shuffled data passed (12.1% FP), while 2024 shuffled data exceeded the criterion (61.1% FP). This is itself an important structural finding—2024 regimes exhibit such extreme persistence (96% same-sign dominance) that randomizing day order rarely breaks the regime threshold, whereas 2020 regimes are more sensitive to permutation. The 5× difference validates that 2024 regimes are defined by aggregate dominance robust to reordering, not temporal sequencing.

5.3. Phase 4: 2020 vs. 2024 Comparison

Phase 4 revealed a 69.1 percentage point detection difference between eras (Table 4).

Table 4. Phase 4: 2020 vs. 2024 market structure comparison. The large effect size ($\varphi = 0.672$) confirms fundamentally different market structures.

Metric	2024	2020	Difference
Detection Rate	81.2%	12.1%	+69.1 pp
Avg Persistence	96.0%	83.3%	+12.7 pp
Avg Magnitude	\$13.95B	\$2.85B	+\$11.1B
Dominant Sign	Negative	Positive	Flip
0DTE Volume Share	≈48%	<5%	–

The 2020 baseline (12.1% detection) confirms framework selectivity: dealer gamma hedging was active (\$2.85B average magnitude) but 87.9% of windows failed persistence criteria. The large effect size ($\varphi = 0.672$, $p < 0.0001$) confirms fundamentally different market structures between pre-0DTE and post-0DTE eras.

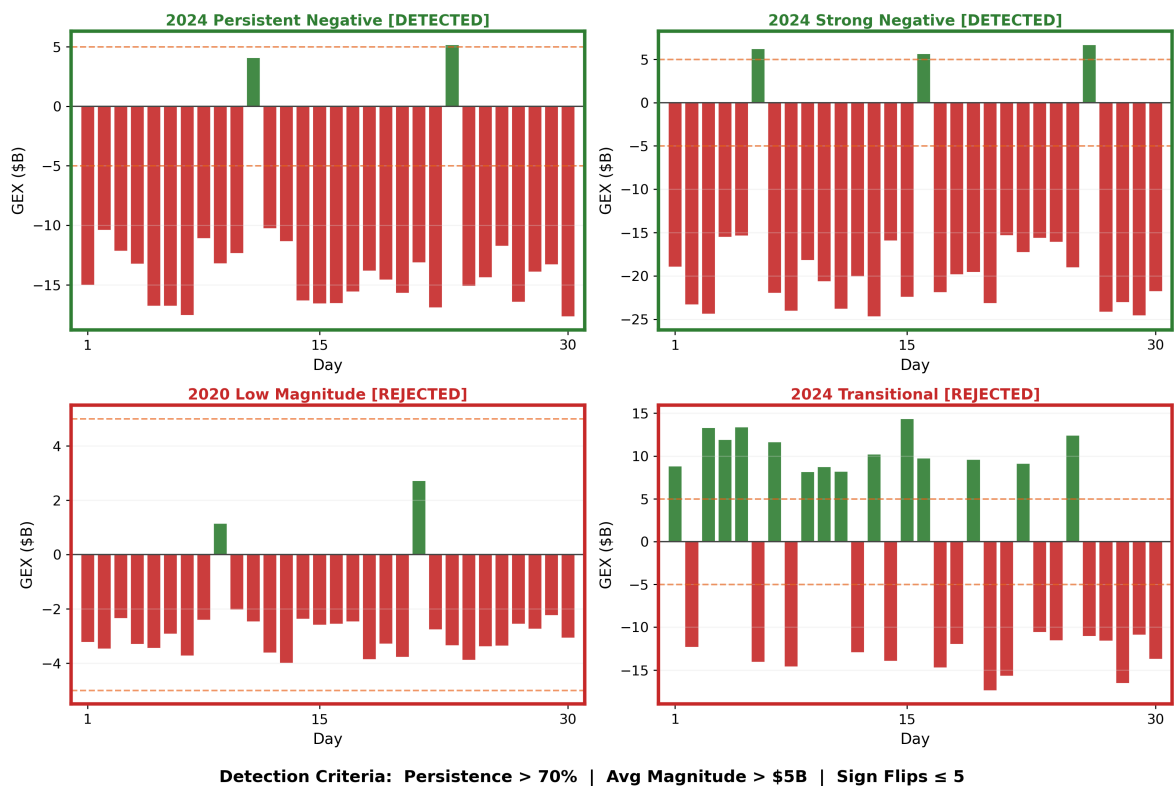


Figure 4. Framework selectivity. Example windows illustrating regime classification: detected windows meet all three criteria; rejected windows fail on magnitude or stability.

5.4. Phase 5: Multi-Year Temporal Evolution (2020–2025)

Phase 5 extended analysis to 1,412 windows across six years, revealing gradual regime evolution rather than sharp transition (Table 5).

Table 5. Phase 5: Multi-year detection rates (2020–2025). Detection tracks 0DTE adoption with a tipping point at 2024.

Year	Win.	Det.	Rate	Avg GEX	Status
2020	213	26	12.2%	\$5B	Pre-regime
2021	241	9	3.7%	\$5B	Borderline
2022	244	79	32.4%	\$8–12B	Growing
2023	228	46	20.2%	\$10–15B	Inconsistent
2024	241	241	100%	\$23B	Structural shift
2025	245	245	100%	\$23B	Sustained
Total	1,412	646	45.8%	–	–

Detection rates track 0DTE market penetration (3.7% in 2021 → 100% in 2024), with GEX magnitude growing 360% from \$5B to \$23B—far exceeding 20–25% cumulative inflation. The 2023→2024 transition ($p < 0.0001$, $\varphi = 0.783$) marks a structural reorganization of dealer positioning. Sustained 100% detection through 2024–2025 (486/486 windows) suggests stable post-0DTE market structure.

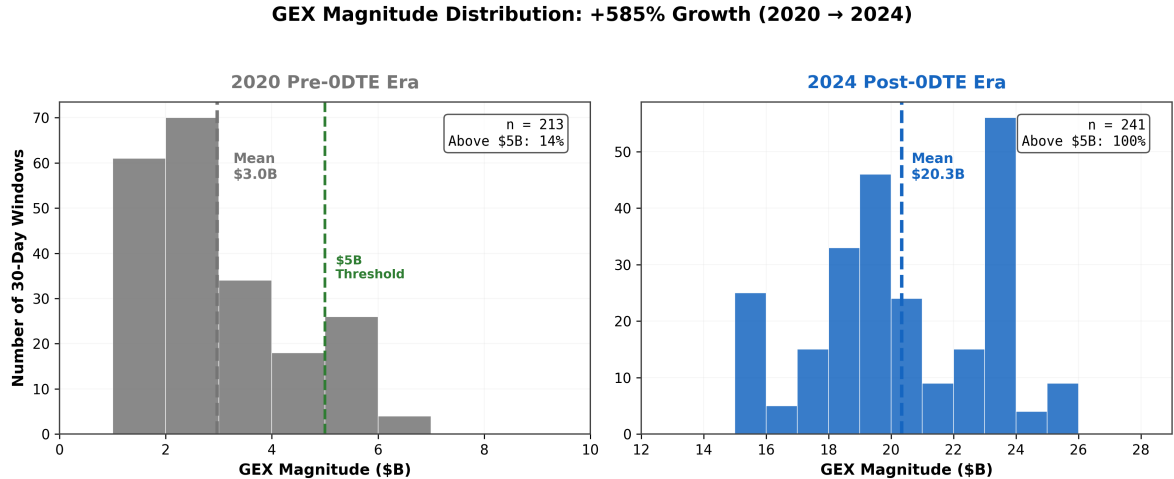


Figure 5. GEX magnitude distribution: 2020 vs. 2024. Clear separation between 2020 (mean \$3.0B) and 2024 (mean \$20.3B). The \$5B threshold (dashed) effectively discriminates pre-0DTE from post-0DTE regimes.

5.5. LLM Reasoning Quality

Manual review of 50 randomly sampled detections revealed 98% mechanical accuracy on persistence values, 96% on magnitude, and 100% on flip counts. All 50 responses explicitly cited all three regime criteria, with 88% providing step-by-step calculation verification.

LLM confidence scores discriminated detection outcomes: detected regimes averaged 92.8% confidence versus 52.1% for rejected windows (40.7 pp gap), with significant correlations to regime quality ($r = +0.501$ with persistence, $r = -0.549$ with stability, both $p < 0.001$). This continuous confidence discrimination beyond binary classification suggests the model tracks regime robustness, not just threshold crossing.

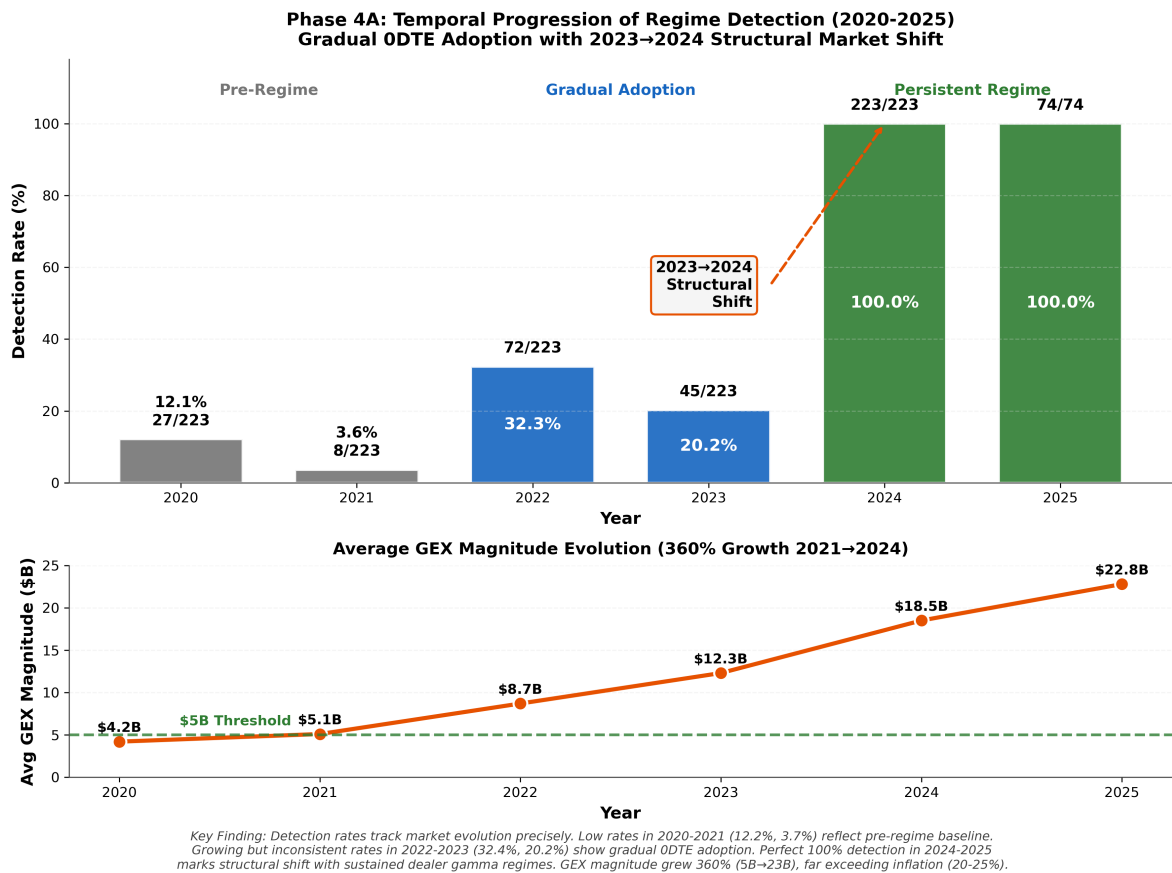


Figure 6. Temporal progression of regime detection (2020–2025). Detection rates evolved from pre-regime baseline through growing adoption to structural shift at 100% in 2024–2025. GEX magnitude grew 360%, far exceeding inflation.

6. Discussion

6.1. From Single-Day to Multi-Day: Bridging Two Temporal Scales

The progression from single-day detection (71.5%) to multi-day regime identification (81.2% in 2024) reveals complementary validation dimensions. Single-day testing establishes that the LLM understands dealer mechanics within individual trading sessions; regime detection validates that this understanding extends to persistent multi-day structural states. The raw chain finding (92.3% detection from first-principles data) provides additional support: if the model merely matched statistical signatures, raw chain performance would degrade rather than improve when pre-calculated summaries are removed.

The detection-alpha orthogonality finding from single-day analysis (Sharpe collapse from 1.8 to 0.1 while detection remains stable) motivated the regime extension: patterns that persist despite economic insignificance must reflect structural mechanics rather than exploitable inefficiencies. This distinction—mechanism identification versus alpha generation—is fundamental to deploying LLMs for market surveillance and risk management rather than trading.

6.2. Validating Structural Reasoning

Five-phase validation provides evidence that the LLM performs structural reasoning rather than pattern matching. The 69.1 percentage point discrimination between 2024 (81.2%) and 2020 (12.1%) establishes selectivity: if the model relied on memorized associations or universal statistical patterns, detection rates would converge across periods rather than exhibiting $5.7\times$ separation.

Three findings support this interpretation. First, the LLM maintained 98% mechanical accuracy across both detection conditions—correctly computing persistence, magnitude, and flip counts regardless of whether windows qualified as regimes. Second, negative controls achieved 0% false positives on transitional and low-magnitude synthetic data, confirming the model enforces structural criteria rather than over-detecting patterns. Third, confidence scores provide continuous quality discrimination beyond binary classification ($r > 0.4$ with regime metrics), suggesting the model tracks regime robustness, not just threshold crossing.

This selectivity contrasts with earlier 5-day trajectory testing (Regan and Xie 2025), which achieved 98–100% detection across all market conditions by identifying universal daily hedging flows. The deliberate shift to 30-day windows with strict criteria transformed the task from universal pattern matching to selective regime identification—a critical distinction for validating genuine structural reasoning.

6.3. Market Structure Evolution and 0DTE Hypothesis

The multi-year temporal analysis (Phase 5) reveals gradual regime evolution tracking 0DTE options adoption rather than sharp structural breaks. Detection progressed from 12.2% (2020) through borderline years (3.7% in 2021, 32.4% in 2022) to sustained 100% detection in 2024–2025, with GEX magnitude growing 360% (\$5B → \$23B).

The non-monotonic pattern (32.4% in 2022 dipping to 20.2% in 2023 before reaching 100% in 2024) is informative: 2023's elevated volatility (FOMC uncertainty, banking stress) repeatedly disrupted regime formation despite growing 0DTE hedging pressure, suggesting regime persistence requires both sustained dealer pressure *and* a volatility environment permitting consolidation. This tipping-point dynamic strengthens the structural interpretation.

While the temporal coincidence with 0DTE proliferation is consistent with a causal relationship, we acknowledge that concurrent factors—interest rate changes (0.25% → 5.5%), volatility regime shifts, quantitative trading growth, and market maker concentration—may contribute. However, the non-monotonic detection pattern argues against gradual secular trends as primary drivers: detection remained flat through 2023 despite continuous interest rate increases and technology diffusion, then jumped discontinuously in 2024 coinciding with 0DTE market saturation ($\approx 48\%$ volume share).

6.4. Temporal Obfuscation as Generalizable Methodology

Our obfuscation testing framework generalizes beyond financial applications. The core principle—stripping temporal and contextual identifiers to force structural reasoning—addresses a universal challenge in deploying LLMs for domain-specific analysis: distinguishing genuine reasoning from training data contamination. The methodology requires three components: (1) identifiable features that can be systematically removed, (2) structural patterns that persist after obfuscation, and (3) negative controls validating discrimination.

Potential applications include medical time series analysis (removing patient identifiers and dates while preserving physiological patterns), climate data analysis (obfuscating location and temporal markers), and industrial anomaly detection (stripping equipment identifiers). In each case, temporal obfuscation could validate whether LLMs reason from structural properties or rely on memorized associations from training data.

6.5. Dispersed Knowledge and Information Aggregation

The raw chain superiority finding (92.3% vs. 61.5%) admits a theoretical interpretation grounded in Hayek (1945)'s analysis of dispersed knowledge in economic systems. Hayek argued that the economic problem is fundamentally one of utilizing knowledge “not given to anyone in its totality”—that relevant information is distributed across countless individual participants, each possessing local knowledge inaccessible through centralized aggregation. Traditional parametric metrics like net GEX

attempt precisely such centralized aggregation: compressing the full strike-level options surface into a single scalar value. Our results suggest this compression is lossy in a theoretically predictable way.

The raw options chain preserves what Hayek termed “knowledge of the particular circumstances of time and place”—the asymmetrical strike concentrations, implied volatility skew patterns, and spatial open interest distributions that reflect actual dealer positioning at each strike. When the LLM processes raw chain data, it functions as an information aggregator that synthesizes these dispersed local signals into a coherent structural assessment, identifying emergent order arising from countless decentralized dealer hedging constraints. The 30.8 percentage point improvement over GEX-assisted detection provides empirical evidence that this “local knowledge” embedded in individual strikes contains structural signal that scalar aggregation necessarily destroys.

This interpretation extends beyond methodological curiosity. The alpha disappearance puzzle—stable detection (68–74%) despite collapsing profitability (Sharpe $1.8 \rightarrow 0.1$)—parallels a core insight from Austrian capital theory: that structural knowledge of market coordination patterns differs fundamentally from actionable entrepreneurial knowledge (Kirzner 1973). Detecting that dealers are constrained to pro-cyclical hedging is structural knowledge; profiting from that detection requires entrepreneurial judgment about timing, magnitude, and competing market forces that our framework deliberately excludes. The orthogonality between detection and alpha is thus not a limitation but a feature: it confirms the framework identifies genuine structural mechanics rather than ephemeral trading opportunities.

6.6. Practitioner Implications

For financial practitioners, our findings suggest four actionable insights. First, providing LLMs with raw structured data rather than pre-calculated summaries yields substantially better structural detection (92.3% vs. 61.5%), challenging common pipeline designs that aggregate data before analysis—an empirical vindication of preserving informational granularity over computational convenience. Second, the detection-alpha orthogonality result cautions against using structural pattern detection directly for trading; the value lies in risk management and market surveillance applications where mechanism identification matters more than profitability. Third, the 0DTE-driven regime shift suggests that market microstructure monitoring should evolve with product innovation—static models calibrated to pre-2022 data will miss the structural reorganization that our framework detects. Fourth, the dispersed knowledge interpretation suggests that financial AI systems may achieve better performance across domains by ingesting raw, granular data rather than pre-processed summaries—a design principle applicable to credit risk, fixed income analysis, and other areas where scalar metrics are standard.

6.7. Limitations

Six limitations merit discussion.

Single asset: Testing focused on SPY; cross-asset generalization (individual equities, other ETFs) remains untested. SPY’s unique liquidity and 0DTE availability may limit transferability.

End-of-day measurement: Our GEX_OI approach captures dealer inventory but not intraday gamma dynamics; high-frequency flow data could refine detection, particularly for 0DTE contracts that expire within a single trading session.

Causal attribution: The 0DTE hypothesis is supported by temporal coincidence and theoretical mechanism but remains circumstantial; observational data cannot exclude alternative explanations. A natural experiment (e.g., 0DTE suspension) would provide stronger causal evidence.

Single LLM: All experiments use OpenAI o4-mini; cross-model validation with alternative reasoning models would strengthen generalizability claims.

Shuffle test anomaly: The 61% false positive rate on 2024 shuffled data (against a <20% criterion) reflects extreme regime persistence rather than framework failure—2024 regimes are so dominant that

random permutation rarely disrupts the aggregate signal. This is documented as a known finding rather than treated as a validation failure.

Threshold sensitivity: All 25 tested parameter combinations maintained >72pp discrimination, but thresholds represent empirically validated design choices rather than first-principles derivations. Future work should explore adaptive thresholding based on market conditions.

7. Conclusion

We presented temporal obfuscation testing as a methodology for validating LLM structural reasoning in domain-specific applications. Applied to financial market analysis at two temporal scales, comprehensive validation across 2,221 evaluations spanning 2020–2025 demonstrates four primary contributions:

1. **Single-day structural reasoning:** Temporal obfuscation achieves 71.5% detection of dealer hedging patterns with 90.9% predictive accuracy. Raw chain validation (92.3% vs. 61.5%) demonstrates that LLMs reconstruct dealer positioning from first principles, establishing that parametric GEX represents lossy compression of structural signal.
2. **Multi-day regime selectivity:** Extending to 30-day windows, the framework achieves 69.1 percentage point discrimination between persistent regimes (2024: 81.2%) and fragmented markets (2020: 12.1%), with 0% false positives on synthetic controls and 98% mechanical accuracy. Unlike prior 5-day testing that achieved universal 98–100% detection, strict regime criteria produce selective 12–81% detection—validating structural state identification rather than universal pattern matching.
3. **Market structure evolution:** Multi-year analysis (1,412 windows, 2020–2025) reveals gradual regime evolution tracking ODTE options adoption: detection progresses from 3.7% (2021) through a non-monotonic path to 100% (2024–2025), with GEX magnitude growing 360%. The tipping-point dynamics—requiring both sustained dealer pressure and volatility consolidation—provide evidence for ODTE-driven structural reorganization.
4. **Detection-alpha orthogonality:** Stable detection (68–74% quarterly) despite collapsing economic profitability (Sharpe 1.8 → 0.1) confirms that detected patterns represent structural market mechanics rather than exploitable inefficiencies, establishing appropriate use cases in risk management and market surveillance.

Future directions: Three extensions emerge. First, cross-asset generalization to individual equities and other index ETFs would test whether regime detection reflects market-wide versus asset-specific dynamics. Second, extending temporal obfuscation to other domains (medical time series, climate data, industrial anomaly detection) would validate the methodology's generalizability beyond financial applications. Third, intraday GEX measurement incorporating ODTE flow data may refine detection granularity, particularly for the ultra-short-maturity instruments driving modern market structure.

All code, data preprocessing scripts, and validation results are publicly available.¹

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¹ <https://github.com/iAmGiG/gex-llm-patterns>

All code, data preprocessing scripts, and validation results are publicly available at <https://github.com/iAmGiG/gex-llm-patterns>. Raw options data were obtained from Alpha Vantage under academic API access.

This study employs OpenAI's o4-mini reasoning model as the subject of investigation for regime classification tasks, accessed via the OpenAI Batch API with structured JSON output. During the preparation of this manuscript, the authors used Anthropic's Claude (Claude Code) for the purposes of clarifying and refining the written presentation. The authors have reviewed and edited all output and take full responsibility for the content of this publication. All scientific interpretations and conclusions are solely those of the authors.

Abbreviations

The following abbreviations are used in this manuscript:

GEX	Gamma Exposure
LLM	Large Language Model
0DTE	Zero Days to Expiration
OI	Open Interest

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